

## Integral monotonicity.

Contrast this with pointwise monotonicity: i.e., a maximum principle holds: e.g.,

$$\partial_t R = \Delta R + 2/R|c|^2$$

↑ nonnegative.

integral quantity is a number instead of a function — obtained by averaging or summing.

Key idea: More chance to hold in complete generality.

# $\mathcal{F}$ -Energy

Genesis: total scalar curvature

Einstein-Hilbert action.

$$E(g) = \int_M R \, d\mu$$

$\propto \text{vol. form}$

Compute 1<sup>st</sup> variation.

$$g(t) \quad \partial_t g = v \text{ symmetric 2-tensor.}$$

Then (let  $V = \text{tr}_g(v)$ ) we have.

$$\partial_t R = -\Delta V + \text{div}(\text{div}(v)) - v \cdot \text{Ric}.$$

$$\partial_t d\mu = -\frac{1}{2} V d\mu.$$

$$\frac{d}{dt} \mathcal{E}(g(t))$$

$$= \int (\Delta V + \operatorname{div}^2(v) - v \cdot \operatorname{Ric} + \frac{1}{2} V R) d\mu$$

$$= \int -v \cdot (\operatorname{Ric} - \frac{1}{2} R g) d\mu.$$

$\swarrow$  Einstein tensor       $\nwarrow$  "undesirable term".

Lesson: RF is not the gradient flow of  $\mathcal{E}$ .

$$(M^n, g, f)$$

$\wedge$  add the "field"  $f$ .

$\mathcal{F}$ -energy.

$$\mathcal{F}(g, f) = \int ( \underset{\substack{\uparrow \\ \text{E-H.}}}{R} + |\nabla f|^2 ) e^{-f} d\mu.$$

$\uparrow$  Dirichlet.

Suppose  $M^n$  and  $g(t)$  and  $f(t)$  satisfy:

$$\frac{\partial}{\partial t} g = -2 \text{Ric} - 2 \nabla^2 f$$

$$\frac{\partial}{\partial t} f = -R - \Delta f$$

similar to RF.



$$\begin{cases} \partial_t g = -2 \operatorname{Ric} - \Delta g \\ \partial_t f = -R - \Delta f = \frac{1}{2} \operatorname{tr}_g (\partial_t g). \end{cases}$$

Exer.:  $\frac{\partial}{\partial t} (e^{-t} d\mu) = 0$

Perelman's monot. formula:

$$\frac{d}{dt} \mathcal{F}(g(t), f(t)).$$

$$= 2 \int_M |\operatorname{Ric} + \nabla^2 f|^2 e^{-f} d\mu.$$

$$\geq 0.$$

There's something deeper here.

Pechman considered.

$(g, f, \tau)$ .

metric for  $\nearrow$  positive  $\#$  time

$\tau$  is a notion of scale.

Function of:

$\mathcal{W}(g, f, \tau)$

$$= \int \left( \tau (R + |\nabla f|^2) + f^{-n} \right) (4\pi\tau)^{-\frac{n}{2}} e^{-f} d\mu.$$

e.g.  $\mathbb{R}^n$ ,  $f = \frac{|x|^2}{4\tau}$

$$\mathcal{W}(g, t, \tau)$$

$$= \int (\tau (R + |\nabla t|^2) + (t - n)/(4\pi\tau))^{-\frac{n}{2}} e^{-t} d\mu.$$

$\mathcal{W}$ -entropy.

$$\frac{\partial}{\partial t} g = -2Ric.$$

$$(*). \quad \frac{\partial}{\partial t} t = -R - \Delta t + |\nabla t|^2 + \frac{n}{2\tau}$$

$$\frac{1}{dt} \tau = -1.$$

Previously

differs change.

$$\frac{\partial}{\partial t} g = -2Ric \quad \underbrace{-2\nabla^2 t}_{\text{differs change}} + \underbrace{L_{\nabla t} g}_{\text{differs change}}.$$

$$\frac{\partial}{\partial t} t = -R - \Delta t + \underbrace{L_{\nabla t} t}_{|\nabla t|^2}.$$

Under (1) for  $g(t)$  (R.T),  $f(t)$  backward HZ  
&  $\tau(t)$ :  $\frac{dE}{dt} = -1$

the W-entropy

$$W(g, f, \tau) = \int (\tau(R + |\nabla f|^2) + f - u) (4\pi\tau)^{-n/2} e^{-t} dv$$

satisfies the following monotonicity  
formula:

$$\frac{d}{dt} W(g(t), f(t), \tau(t)).$$

$$= 2\tau \int_M |Ric + \nabla^2 f - \frac{1}{2\tau} g|^2 (4\pi\tau)^{-\frac{n}{2}} e^{-t} dv.$$

$$\geq 0. \quad \text{Proof: calculation.}$$

The eqns + might seem arbitrary.

Motivation:

$$RF + HE.$$

$$\begin{cases} \frac{\partial}{\partial t} g = -2Ric \end{cases}$$

$$\begin{cases} \frac{\partial}{\partial t} u = \Delta u, \text{ where } \Delta = \Delta_{g(t)}. \end{cases}$$

Heat operator  $\square = \partial_t - \Delta$ .

Adjoint/conjugate heat operator:

$$\square^* = -\partial_t - \Delta + R.$$

$$\int_I \int_M b \square a \, d\mu_{g(t)} \, dt = \int_I \int_M \square^* b \cdot a \, d\mu_{g(t)} \, dt.$$

Define  $u = (4\pi\tau)^{-n/2} e^{-t}$ .

Calculation:  $\square^* u = 0$ .

Instead of  $(M, g, t, \tau)$ .

consider, equivalently,  $(M, g, u, \tau)$ .

heat kernel associated to RF  
is very fundamental.

Exer:  $\frac{d}{dt} \int u d\mu = 0$  under (\*).

so  $\int u d\mu = \text{const.}$  ..

Given  $g$  &  $\tau$  :

$$\inf_{\int u d\mu = 1} \mathcal{W}(g, \tau, \tau) = \mu(g, \tau) > -\infty$$

$\uparrow$   
log Sabotage  
req..

Exm:  $\frac{d}{dt} \mu(g(t), \tau(t)) \geq 0.$

So. if  $t_1 \leq t_2$ , then.

$$\mu(g(t_2), \tau(t_2)) \geq \mu(g(t_1), \tau(t_1)).$$

Let  $T \in (0, \infty).$

$$\mathcal{V}_T(g) = \inf_{t \in (0, T]} \mu(g, \tau) > -\infty.$$

So we have an integral monovary.

This has an analysis flavor.

$$\mathcal{W}(g, A, \pi) = \int_M \left( \frac{1}{2} (R + |Df|^2) + f - n \right) (4\pi\tau)^{-\frac{n}{2}} e^{-f} d\mu.$$

not so important  
for a specific application

key aspect of  $\mathcal{W}$   
is

input

$$\int f e^{-f} \sim \text{Nash entropy } \mathcal{N}$$

which has certain advantages,

depending on your perspective,

over  $\mathcal{W}$ . Let  $w = e^{-f}$

$$\int f e^{-f} = \int -\log w \cdot w$$

Entropy



Geomet:- application of  $\mathcal{V}$  monot.

Let  $(x_0, t_0) \in M \times [0, T]$  &  $0 \leq r \leq \rho$ .

$\uparrow$                        $\uparrow$   
 pre scale            upper

$$\mu(g(t_0), r^2) \stackrel{\text{monot}}{\geq} \mu(g(0), r^2 + t_0) \quad \text{bound.}$$

$$\text{Now } 0 \leq r^2 + t_0 \leq \rho^2 + T.$$

$$\geq \mathcal{V}_{\rho^2+T}(g(0)) \stackrel{\text{def}}{=} -C(g(0), T, \rho) = -C.$$

We have

$$\mathcal{V}(g(t_0), t, r^2) \geq -C$$

$$\forall t \text{ s.t. } \int u \, d\mu = 1.$$

We have.

$$-c \leq \mathcal{W}(g(t, \cdot), t, r^2).$$

$$= \int (r^2 (R + |\nabla \phi|^2) + t - n) u \, dy.$$

Test function for  $(x_0, t_0)$ . at space  
scale  $r$ :

$u = c \cdot \chi_B$ , characteristic fcn

of  $B = B_r(x_0, t_0) = B_r^{g(t_0)}(x_0)$ .

where  $c$  is determined by the normalization

$$1 = \int u = c \operatorname{Vol} B. \Rightarrow c = \frac{1}{\operatorname{Vol} B}.$$

$$-C \leq \mathcal{W}$$

$$= \int \underbrace{r^2(R + |\nabla \phi|^2)}_{\text{ignore (a lie)}} + 4 - n \big) u \, d\mu.$$

ignore (a lie).

put condition:

$$\boxed{r^2 R \leq 1 \text{ on } B = B_r(x_0, t_0).$$

$$-C \leq \mathcal{W} \leq C + \underbrace{\int f u \, d\mu}_M. \quad C = \frac{1}{\text{vol } B}.$$

$$u = 0 \Rightarrow f u = 0.$$

$$u = \varepsilon \Rightarrow 0.$$

$$\simeq -\ln \varepsilon \cdot \varepsilon \rightarrow 0.$$

so just consider.

B.

Recall.

$$-C \leq \int_B f u \, d\mu.$$

$$u = c \quad \text{i.e.} \quad e^{-f} = c(4\pi r^2)^{n/2} \\ \text{in } B.$$

$$\text{So} \quad -f = \log c + \log(r^n) + c(n).$$

$$-C \leq \int_B f u \, d\mu = f.$$

$$= -\log c - \log(r^n) + c(n).$$

So.

$$-C \leq \log \frac{V(B_r(x_0, t_0))}{r^n}.$$

$$\Rightarrow \frac{V(B_r(x_0, t_0))}{r^n} \geq K = K(g/o, T, \rho) > 0. \quad \underline{\underline{NLC}}.$$